

VAJRA — KAVACH-AI's Custom Fine-Tuned Specialist Model

Team briefing document. Padho poora — kisi ko bhi Vajra ke baare me koi bhi sawaal pooch sakta hai.

1. Humne Vajra Kyun Banaya (The Use Case)

Llama 3.1 ek "**Generalist**" hai — ek Principal Engineer jaisa jisko duniya bhar ki knowledge hai, lekin chalne me zyada time aur GPU/RAM lagta hai. Vajra ek "**Specialist**" hai — ek Expert Clerk/Technician jaisa jisko puri duniya ki knowledge nahi hai, sirf 3 cheezein bahut achhi tarah aati hain:

1. Data ko sahi format me present karna (bullet points + citation)
2. Exact source citation likhna (document, page, revision)
3. Agar plant me wo machine/asset exist hi nahi karta, seedha bol dena — guess nahi karna

Fayda (asli reasoning)

Agar har chhote, structured sawaal ke liye bhi bade Llama 3.1 ko jagayenge, system slow ho jayega aur GPU resources waste honge. Isliye chhote/repetitive lookups Vajra handle karta hai (fast, CPU pe), aur jab genuinely complex multi-step reasoning ya troubleshooting chahiye, tabhi Llama 3.1 (GPU, heavy) call hota hai. Refinery jaisi jagah par ~80% queries repetitive/structured hote hain, ~20% genuinely complex — isliye ye split real-world traffic pattern ke hisaab se design kiya gaya hai.

2. Vajra Actual Mein Hai Kya (Deep Learning vs ML, aur QLoRA)

ML (Random Forest, SVM) nahi use kiya — kyunki humein natural language text samajhna aur generate karna hai, jo classical ML algorithms nahi kar sakte.

Scratch se naya Deep Learning model bhi nahi banaya — ek naya LLM pretrain karne me millions of dollars aur mahino ka compute lagta hai; ye kisi college team ke scope me nahi hai.

Jo actually kiya: Ek chhota pre-trained open-source Deep Learning model liya (Transformer-based LLM — Llama-3.2-3B / Qwen-2.5-1.5B class), jisko English aur basic reasoning pehle se aati hai, aur uske upar apna proprietary refinery data inject kiya **QLoRA (Quantized Low-Rank Adaptation)** technique se.

QLoRA — do hisson me

Q (Quantized): Model ke bade size ko 4-bit precision me sikhod deta hai, taaki wo 6GB RTX 4050 GPU me fit ho jaye — normal full-precision training ke liye 100GB+ RAM chahiye hoti.

LoRA (Low-Rank Adaptation): Poore model ko dobara train karne ke bajaye, model ke upar ek chhoti "adapter" layer lagayi jaati hai aur SIRF usi chhoti layer ko train kiya jaata hai — baaki model ke original weights freeze rehte hain. Isse training bahut tez aur halki ho jaati hai.

Is technique se poora training **Piyush ke laptop pe hi, offline, koi bahar ka data liye bina** hua — ye khud sovereignty principle ke saath consistent hai.

3. Training Dataset — 4 Categories

Vajra ko sirf humara diya hua data hi seekhne ko mila hai. Dataset 4 categories me structured hai:

#	Category	Example
1	Direct Factual Q&A + Citation (40%)	Q: "BOILER-B101 ka max operating temp kya hai?" A: "450 degC (Source: Maintenance Log Rev-4, Page 2)"
2	CSV / Numeric Summary (20%)	Q: "PUMP-A ka pichla maintenance kab hua?" A: "2026-03-14, 3 failures reported."
3	Abstention / Safety Fallback (20%) -- SABSE IMPORTANT	Q: "REACTOR-999 ka pressure kitna hai?" A: "[INSUFFICIENT EVIDENCE] Database me koi jankari nahi hai. Human approval zaroori hai."
4	Conflict Detection (20%)	Q: "Boiler Rev-2 aur Rev-4 me farak?" A: "[CONFLICT ALERT] Rev-2: 400 degC, naye Rev-4 me: 450 degC."

Category 3 (Abstention) sabse important hai — isi se Vajra **khud** hallucination rokna seekhta hai, sirf prompt-engineering pe depend nahi rehta.

4. Switching Kaise Kaam Karta Hai (Dual-Track Router)

System ko ye rule-based logic se pata chalta hai ki kaunsa query Vajra ko jaaye, kaunsa Llama 3.1 ko:

Query Type	Kya Detect Hota Hai	Kahan Route Hota Hai
"What is thermodynamics?" / "Write Python code"	Koi specific asset-tag nahi (general query)	Llama 3.1 (general reasoner)
"VALVE-12 ka status batao"	Specific asset-tag pattern (VALVE-12) detect hua	Vajra (specialist, citation-enforced)
Image / P&ID upload	Visual input	Qwen2.5-VL (Vaibhav's laptop)
CSV / calculation request	Structured data / calc keywords	Qwen2.5-Coder (Vaibhav's laptop)

Important — ye "hot-swap" nahi hai

Piyush ke laptop pe Llama 3.1 (8B) already ~5.5GB VRAM leta hai 6GB GPU pe — agar Vajra bhi GPU pe load karte, OOM crash hota. Isliye **Vajra ko CPU-only force kiya gaya hai** (Ollama Modelfile me `PARAMETER num_gpu 0`). Result: Llama GPU pe, Vajra CPU pe -- dono ek saath available hain, koi unload/reload wait nahi karna padta. Router sirf decide karta hai mode l field me kya bhejna hai -- ye ek simple if-else decision hai, koi hardware-level switch nahi.

Zaroori rule: Agar Vajra abstain karta hai ("[INSUFFICIENT EVIDENCE]"), system usko chup-chaap Llama ko retry nahi karta — seedha human review ke liye escalate hota hai. Isse abstention guarantee kabhi bypass nahi

hoti.

5. "Better Result" — Kaise Prove Karte Hain (claim nahi, proof)

Demo me hum base Llama 3.1 aur Vajra ko **same query** live bhejte hain, side-by-side, aur automatically compare karte hain:

- **Latency** — Vajra kitna tez respond karta hai (CPU pe bhi, chhota model hone ki wajah se)
- **Citation format** — kya response me "(Source: ..., Page ..., Rev-...)" pattern hai — regex se check hota hai
- **Abstention correctness** — unknown asset (jaise REACTOR-999) poochne par kya model sahi se "insufficient evidence" bolta hai, ya guess karta hai

Judge ke saamne ye dikhega

Base Llama 3.1: generic, hedge-y jawab, citation format follow nahi karta.

Vajra (fine-tuned): direct, structured jawab, source/page/revision ke saath, aur unknown asset pe sahi se abstain karta hai.

Ye numbers ke saath live dikhta hai — sirf "humara model better hai" bolna nahi, balki ek script dono ko run karke result print karta hai.

6. "Ye Data Toh Random/Fake Hai" — Iska Jawab

Judges ko pata hota hai ki kisi student team ke paas real ONGC/IOCL ka confidential data nahi ho sakta — wo highly confidential aur illegal hota hai kisi ke paas hona. Isliye hum isko **"false info"** nahi balki **"Proof of Concept (PoC) with Synthetic/Mock Data"** bolte hain.

Exact jawab jo judge ko dena hai

"Sir, due to industrial data privacy and confidentiality, we obviously cannot use real telemetry or P&ID data from a real plant. Instead, we have seeded our database and trained Vajra using Synthetic (Mock) Data.

Our goal today is not to give you the real temperature of a real boiler. Our goal is to demonstrate the ARCHITECTURE and CAPABILITY -- how the system routes queries, how it enforces citations, how it detects conflicts, and how it safely abstains from hallucinating.

Tomorrow, if a refinery deploys this system, they simply replace our demo data with their actual real-time plant logs. The entire AI mesh works exactly as demonstrated today, without needing internet."

Core insight: Humne model ko "jankari" (information) nahi sikhayi hai, humne usko "tareeka" (behavior aur format) sikhaya hai. Data fake ho sakta hai, lekin **system jis tarah se us data ko handle karta hai** (citation nikalna, jhooth na bolna, conflict detect karna) — wo capability 100% real hai. Data replace karte hi system enterprise-ready ho jaata hai.

7. Vajra-Specific Q&A (rapid fire)

Q: Jab Llama 3.1 hai, Vajra ki kya zarurat hai?

A: Llama generalist hai (heavy, slow, sab kuch samajhta hai). Vajra specialist hai (light, fast, sirf structured lookups + citation + abstention). Har chhote sawaal ke liye bade model ko jagana wasteful hai.

Q: System ko kaise pata router use karna hai konsa model?

A: Rule-based dual-track router: agar query me specific asset-tag (jaise VALVE-12) hai -> Vajra. Agar general/open-ended reasoning hai -> Llama 3.1. Image -> Vision model. CSV/calc -> Coder model.

Q: Vajra 'special' kyun hai, sirf ek chhota model toh hai?

A: Base pre-trained models (bina fine-tune ke) unknown facts pe hallucinate karte hain. Humne Vajra ki 'brain surgery' ki -- QLoRA se usko strict rules sikhaye: evidence na ho toh guess mat karo, hamesha citation do. Yahi trained BEHAVIOR hi uski uniqueness hai, sirf model size nahi.

Q: QLoRA normal fine-tuning se alag kaise hai?

A: Normal fine-tuning poore model ke saare weights update karta hai -- bahut RAM/GPU chahiye. QLoRA model ko 4-bit tak quantize (chhota) karta hai aur sirf ek chhoti adapter layer train karta hai -- isliye 6GB laptop GPU pe bhi chal jaata hai.

Q: Agar Vajra abstain kare, system Llama ko retry karta hai?

A: Nahi -- jaan-bujh kar nahi karta. Abstention seedha human review ke liye jaata hai. Agar hum silently bade model ko retry karte, hallucination-prevention guarantee hi bypass ho jaati.

Q: Vajra kaunsa laptop pe, GPU ya CPU pe chalta hai?

A: Piyush ke laptop pe, lekin CPU-only mode me force kiya gaya hai (Ollama Modelfile: `PARAMETER num_gpu 0`), taaki wo Llama 3.1 (jo GPU pe already ~5.5GB VRAM use karta hai) ke saath resource conflict na kare.

Ek line jo yaad rakhni hai

"Vajra ne koi nayi jaankari nahi seekhi -- usne sahi TAREEKA seekha hai: citation dena, conflict pakadna, aur jab pata na ho toh 'pata nahi' bolna. Yahi wo cheez hai jo isko har doosri team ke generic chatbot se alag banati hai."